

The screenshot displays the MongoDB Cloud Portal interface. On the left, a sidebar contains navigation links: 'My Queries', 'Data Modeling', and 'CLUSTERS (1)'. Below these is a search bar for clusters and a list of clusters including 'Cluster0'. Under 'Cluster0', there are sub-clusters like 'admin', 'local', 'rag_ingestion', 'testcases', 'userstories', and 'sample_mflix'. The 'testcases' cluster is selected, and its details are shown in the main area. The main area has a top bar with 'testcases' and a '+' icon. Below this, there's a breadcrumb 'Cluster0 > rag_ingestion > testcases'. A tab bar shows 'Documents 6.4K', 'Aggregations', 'Schema', 'Indexes 1', and 'Validation'. The 'Documents' tab is active, showing a list of documents. The first document is expanded, displaying a JSON document with the following content:

```
{
  "_id": "ObjectID('6aa96c657d0b5f468301091')",
  "module": "Registration",
  "id": "TC_005",
  "prerequisites": "PRN/ERN Number",
  "title": "Conversion from PRN/ERN Reg to UHID",
  "description": "Verify Conversion from PRN/ERN Reg to UHID",
  "steps": [
    "1. Launch the application",
    "2. Login to the application with valid creden..."
  ],
  "expectedResults": "Should get convert successfully from PRN/ERN Reg to UHID",
  "automationManual": "Automated",
  "priority": "P1",
  "createdBy": "Prabakaran",
  "createdAt": "45165",
  "lastModified": "45176",
  "linkedStories": "HC-080"
}
```

At the bottom of the interface, there's a status bar with 'Alert System Status: All Good' and copyright information for MongoDB, Inc. The footer also includes links for 'Status', 'Terms', 'Privacy', 'Atlas Blog', and 'Contact Sales'.

Now retrieval process

Retrieval Pipeline:

Query Preprocessing

Vector search

BM25 search

Hybrid search

Reranking

Deduplication

Summarization

I

Context ->LLM -->generate the new testcases

End to end Pipeline.

Before that BM25 search created

The top screenshot shows the MongoDB Atlas Search & Vector Search interface. The left sidebar lists navigation options: Search Indexes, AUTO EMBEDDING, NATIVE RERANKING, and SHORTCUTS. The main content area displays a table of search indexes. A yellow banner at the top indicates that the 3 search and vector index limit for the M0 free tier cluster has been reached, with a link to 'Upgrade Cluster'.

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)
rag_ingestion	testcases	bm25_index	READY	✓	search	"automationManual", "createdBy", "description", "expectedResults", "id", ...	6,354 (100%) indexed of 6,354
rag_ingestion	testcases	vector_index_testcase	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description", ...	6,354 (100%) indexed of 6,354
rag_ingestion	userstories	vector_index_userstory	READY	✓	vectorSearch	"embedding", "jiraMetadata.projectKey", "status.name", "priority.name", "assignee.displayName", ...	190 (100%) indexed of 190

The bottom screenshot shows a VS Code editor with the .env file for the rag-mongo-demo_E2E project. The file contains configuration for MongoDB Atlas, search indexes, and LLM models.

```
1 # ===== MongoDB Database & Collections =====
2 # MongoDB Atlas Connection URI with credentials
3 MONGODB_URI="mongodb://muktigarg81_db_user:C5pj9f1H5K0X#Evi@ac-1fstggeg-shard-00-00.xpros31.mongod
4 # Database Name
5 DB_NAME="rag_ingestion"
6
7 # Test Cases Collection & Search Indexes
8 COLLECTION_NAME="testcases"
9 VECTOR_INDEX_NAME="vector_index_testcase"
10 BM25_INDEX_NAME="bm25_index"
11
12 # User Stories Collection & Search Indexes
13 USER_STORIES_COLLECTION_NAME="userstories"
14 USER_STORIES_VECTOR_INDEX_NAME="vector_index_userstory"
15 USER_STORIES_BM25_INDEX_NAME="bm25_user_stories"
16
17 # ===== Task-Specific LLM Models =====
18 # 1. Primary Test Case Generation / QA LLM (groq | gemini | openai | claude | mistral | nvidia)
19 LLM_PROVIDER="groq"
20 LLM_MODEL="gemma2-9b-it-groq"
```

The terminal output shows the server restarting and running on port 3001, with the API available at http://localhost:3001/api.

Query Preprocessing is three step process

1. Normalization
2. Abbreviation expansion
3. Synonyms expansion

The screenshot displays the RAG Pipeline Enterprise v2.0 interface. The left sidebar contains a navigation menu with sections for DATA PIPELINE (Convert to JSON, Embeddings & Store) and SEARCH ENGINES (Query Preprocessing, Vector Search, BM25 Search, Hybrid Search, Score Fusion). The main content area is titled 'RAG Pipeline > Query Preprocessing' and features a 'Preprocess Query' button. Below this, the 'Query Transformation Pipeline' is shown with three steps: 1. ORIGINAL QUERY: 'TC_027 merge UHID records'; 2. NORMALIZATION: 'tc 027 merge uhid records' (5 tokens); 3. ABBREVIATION EXPANSION: 'test case 027 merge unique health id records' (2 expansions). The interface is running on http://localhost:3000.

Product managem...

Post Attendee - Zoo...

Google Calendar ~...

IRONLADY

SDET

TestLeaf Program

AIAgents

Phase2 TestLeaf

SeleniumTestLeaf

yoga class

Expanded:

RAG Pipeline

Query Preprocessing

Transform & expand search queries

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DATA PIPELINE

Convert to JSON

Embeddings & Store

SEARCH ENGINES

Query Preprocessing

Vector Search

BM25 Search

Hybrid Search

Score Fusion

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1

SYNONYM EXPANSION

5 variations

Multiple query variations generated for comprehensive search coverage

1. "TC_027 test case 027 merge unique health id records"

2. "TC_027 examination case 027 merge unique health id records"

3. "TC_027 investigation case 027 merge unique health id records"

+ 2 more variations

PROCESSING METADATA

Processing Time	Token Count	Abbreviations	Synonyms
2ms	5	2	2

1

HOW QUERY PREPROCESSING WORKS

1. Normalization:

Converts text to lowercase, removes special characters, extracts test case IDs

2. Abbreviation Expansion:

Expands healthcare abbreviations (UHID → unique health id, OTP → one time password)

3. Synonym Expansion:

Generates multiple query variations using synonyms (patient → customer/user/individual)

4. Search Execution:

Uses all variations to perform comprehensive search with better recall

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Ask Gemini

http://localhost:3000

Product managem... Post Attendee - Zoo... Google Calendar ~... IRONLADY SDET TestLeaf Program AIAgents Phase2 TestLeaf SeleniumTestLeaf yoga class All Bookmarks

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DATA PIPELINE

Convert to JSON

Embeddings & Store

SEARCH ENGINES

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1

Query Preprocessing

Transform your query through normalization, expansion, and synonym generation.

Enter Your Query

TC_027 merge UHID records

Quick Examples:

UHID patient login issue OTP not working

TC_027 merge UHID records

doctor appointment booking IP admission

password reset OTP verification

ER patient registration BP monitoring

Preprocess Query

1

HOW QUERY PREPROCESSING WORKS

1. Normalization:

Converts text to lowercase, removes special characters, extracts test case IDs

2. Abbreviation Expansion:

Expands healthcare abbreviations (UHID → unique health id, OTP → one time password)

3. Synonym Expansion:

Generates multiple query variations using synonyms (patient → customer/user/individual)

4. Search Execution:

Uses all variations to perform comprehensive search with better recall

Pipeline Steps

Abbreviations

Synonym Variations

Abbreviation Mappings

Abbreviation	Full Form	Position
UHID	unique health id	13
TC	test case	-

BEFORE & AFTER COMPARISON

BEFORE (with abbreviations)

tc 027 merge uhid records

AFTER (expanded)

test case 027 merge unique health id records

Generated Query Variations (5)

These variations will be used to perform comprehensive search across the database

Variation 1

Primary



"TC_027 test case 027 merge unique health id records"

Variation 2



"TC_027 examination case 027 merge unique health id records"

Variation 3



"TC_027 investigation case 027 merge unique health id records"

Variation 4



"TC_027 analysis case 027 merge unique health id records"

Variation 5



"TC_027 screening case 027 merge unique health id records"

SYNONYM MAPPINGS (2)

The screenshot displays the RAG Pipeline web application interface. The browser address bar shows the URL `http://localhost:3000`. The application has a sidebar on the left with a 'DATA PIPELINE' section containing 'Convert to JSON' and 'Embeddings & Store', and a 'SEARCH ENGINES' section with 'Query Preprocessing' (highlighted), 'Vector Search', 'BM25 Search', 'Hybrid Search', and 'Score Fusion'. At the bottom of the sidebar is a button for 'Enterprise AI Stack' with subtext 'MongoDB Atlas • Mistral AI • Groq'. The main content area is titled 'RAG Pipeline > Query Preprocessing' and includes a 'Transform & expand search queries' button. It displays three query variations (3, 4, and 5) with their respective copy and search icons. Below these is a 'SYNONYM MAPPINGS (2)' table:

Term	Synonyms	Position
test	examination investigation analysis screening	0
merge	combine join unite consolidate integrate	3

Vector Search

The screenshot displays the RAG Pipeline | Testleaf Enterprise v2.0 interface. The left sidebar contains navigation options under 'SEARCH ENGINES' (Query Preprocessing, Vector Search, BM25 Search, Hybrid Search, Score Fusion) and 'AI PROCESSING' (Summarize & Dedup, Prompt & Schema). The main area shows a 'Vector Search' query for 'TC_027 merge UHID records'. Below the query, there are 'TRY AN EXAMPLE QUERY' suggestions and 'Advanced Metadata Filters'. The search results are displayed in a table format.

Search Query: "TC_027 merge UHID records"
Performance: \$0.000001 cost, 13 tokens

Search Results
3 results ranked by semantic similarity

Test Case	Similarity	Description	Source
MERGE UHID	93.1%	Verify user able to Merge Master UHID and Child UHID.	converted-178...
Registration			
DEMERGE UHID	91.1%	Verify user able to Demerge Master UHID and Child UHID.	converted-178...
Registration			
CONVERTED TO UHID	89.9%	Verify Converted to UHID report	converted-178...
Registration			

BM25 search

The screenshot displays the RAG Pipeline interface for a BM25 search. The left sidebar shows the 'RAG Pipeline Enterprise v2.0' logo and a navigation menu with 'SEARCH ENGINES' (Query Preprocessing, Vector Search, BM25 Search, Hybrid Search, Score Fusion) and 'AI PROCESSING' (Summarize & Dedup, Prompt & Schema). The main content area is titled 'RAG Pipeline > BM25 Search' and features a 'Keyword-based search algorithm' toggle. A search bar contains the query 'TC_027 merge UHID records' and a 'Results Limit' of 3. A red 'Search' button is visible. Below the search bar, a blue banner indicates 'BM25 Search Results: Found 3 test cases in 259ms' with 'Search Type: bm25'. The 'Search Results (3)' section shows the first result, '#1 TC_027' with a 'Score: 68.36'. The result details include 'Merge UHID', 'Module: Registration', 'P1' priority, 'Risk: Low', and 'Automated' status. The description states: 'Verify user able to Merge Master UHID and Child UHID.' The steps are: '1. Launch the application 2. Login to the application with valid credentials. 3. Click on Patient Service Tab and select Registration from the drop down. 4. Functions->Transaction->Merge/Demerge. 5. Enter Master UHID and Click on Go option. 6. Enter Child UHID and Click on Go option. 7. Click on Merge UHID'. The expected results are: 'Selected UHID Merged Successfully with Master UHID Message should be displayed'. The source is 'converted-1789826731698.json' and it was created on '9/19/2026'. The second result, '#2 TC_059' with a 'Score: 42.31', is partially visible below.

BM25 Keyword Search
Fast keyword-based search using BM25 algorithm. Best for exact matches and term queries.

Search Query: TC_027 merge UHID records | Results Limit: 3 | Search

Show Filters

BM25 Search Results: Found 3 test cases in 259ms
Search Type: bm25

Search Results (3)

#1 TC_027 Score: 68.36

Merge UHID

Module: Registration | P1 | Risk: Low | Automated

Description: Verify user able to Merge Master UHID and Child UHID.

Steps: 1. Launch the application 2. Login to the application with valid credentials. 3. Click on Patient Service Tab and select Registration from the drop down. 4. Functions->Transaction->Merge/Demerge. 5. Enter Master UHID and Click on Go option. 6. Enter Child UHID and Click on Go option. 7. Click on Merge UHID

Expected Results: Selected UHID Merged Successfully with Master UHID Message should be displayed

Source: converted-1789826731698.json | Created: 9/19/2026

#2 TC_059 Score: 42.31

Merge/Demerge report

Module: Registration | P1 | Risk: Low | Automated

Description: Verify Merge/Demerge report

Steps: 1. Launch the application 2. Login to the application with valid credentials. 3. Click on Patient Service Tab and select Registration from the drop down. 4. Navigate to Functions->Reports->Merge Demerge report 5. Provide the input details Uhid/ From Date and To Date and click on report and observe the report 6. Click on Export to Excel button (if required)

Expected Results: Report Merge/Demerge report should be generated

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http://localhost:3000

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SEARCH ENGINES

Query Preprocessing

Vector Search

BM25 Search

Hybrid Search

Score Fusion

AI PROCESSING

Summarize & Dedup

Prompt & Schema

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RAG Pipeline > Tr BM25 Search

Keyword-based search algorithm

#2 TC_027 Score: 42.31

Merge/Demerge report

Module: Registration P1 Risk: Low Automated

Description: Verify Merge/Demerge report

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Registration from the drop down. 4.Navigate to Functions->Reports->Merge Demerge report 5.Provide the input details Uhid/ From Date and To Date and click on report and observe the report 6.Click on Export to Excel button (if required)

Expected Results: Record Merge or Demerge request report should get generate

Source: converted-1789826731698.json Created: 9/19/2026

#3 TC_028 Score: 35.49

Demerge UHID

Module: Registration P1 Risk: Low Automated

Description: Verify user able to Demerge Master UHID and Child UHID.

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Registration from the drop down. 4.Functions->Transaction->Merge/Demerge. 5.Enter Master UHID and Click on Go option. 6. Click on Demerge Consent and Click on Demerge UHID

Expected Results: Selected UHID Demerged Successfully From Master UHID Message should be displayed

Source: converted-1789826731698.json Created: 9/19/2026

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RAG Pipeline > Hybrid Search

Combined BM25 + Vector search

Hybrid Search

Combines BM25 keyword matching with semantic vector search for best results.

Search Query TC_027 merge UHID records Results Limit 2 Search

Search Weights

BM25 Weight (Keyword): 50% Vector Weight (Semantic): 50%

0% 50% 100% 0% 50% 100%

Balanced (50/50) Keyword Heavy (70/30) Semantic Heavy (30/70)

Show Filters

Hybrid Search Results: Found 2 test cases in 866ms

Stats: 2 in both indexes,0 BM25 only,0 vector only

Timing: BM25: 108ms, Vector: 751ms

Cost: \$0.000001 (13 tokens)

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DATA PIPELINE

- Convert to JSON
- Embeddings & Store

SEARCH ENGINES

- Query Preprocessing
- Vector Search
- BM25 Search
- Hybrid Search
- Score Fusion

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RAG Pipeline > Hybrid Search

Combined BM25 + Vector search

Search Query: TC_027 merge UHID records Results Limit: 2

Search

Search Weights

BM25 Weight (Keyword): 50% Vector Weight (Semantic): 50%

0% 50% 100% 0% 50% 100%

Balanced (50/50) Keyword Heavy (70/30) Semantic Heavy (30/70)

Show Filters

Hybrid Search Results: Found 2 test cases in 793ms

Stats: 2 in both indexes, 0 BM25 only, 0 vector only

Timing: BM25: 282ms, Vector: 503ms

Cost: \$0.000001 (13 tokens)

Search Results (2)

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DATA PIPELINE

- Convert to JSON
- Embeddings & Store

SEARCH ENGINES

- Query Preprocessing
- Vector Search
- BM25 Search
- Hybrid Search
- Score Fusion

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RAG Pipeline > Hybrid Search

Combined BM25 + Vector search

Search Results (2)

#1 TC_027 Hybrid: 96.5% Found in: both

Merge UHID

Module: Registration P1 Risk: Low Automated

Description: Verify user able to Merge Master UHID and Child UHID.

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Registration from the drop down. 4.Functions->Transaction-->Merge/Demerge. 5.Enter Master UHID and Click on Go option. 6.Enter Child UHID and Click on Go option. 7.Click on Merge UHID

Expected Results: Selected UHID Merged Successfully with Master UHID Message should be displayed

Score Breakdown:

BM25: 100.0% (raw: 68.36) Vector: 93.1% (raw: 0.93) Hybrid: 96.5%

Source: converted-1789826731698.json Created: 9/19/2026

#2 TC_028 Hybrid: 71.5% Found in: both

Demerge UHID

Module: Registration P1 Risk: Low Automated

Description: Verify user able to Demerge Master UHID and Child UHID.

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Registration from the drop down. 4.Functions->Transaction-->Merge/Demerge. 5.Enter Master UHID and Click on Go option. 6. Click on Demerge Consent and Click on Demerge UHID

Expected Results: Selected UHID Demerged Successfully From Master UHID Message should be displayed

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How can I create a new folder

Ask Gemini

http://localhost:3000

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DATA PIPELINE

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RAG Pipeline

Hybrid Search

Combined BM25 + Vector search

Hybrid Search Results: Found 2 test cases in 793ms

Stats: 2 in both indexes,0 BM25 only,0 vector only

Timing: BM25: 282ms, Vector: 503ms

Cost: \$0.000001 (13 tokens)

Search Results (2)

#1 TC_027 Hybrid: 96.5% Found in: both

Merge UHID

Module: Registration P1 Risk: Low Automated

Description: Verify user able to Merge Master UHID and Child UHID.

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Registration from the drop down. 4.Functions->Transaction-->Merge/Demerge. 5.Enter Master UHID and Click on Go option. 6.Enter Child UHID and Click on Go option. 7.Click on Merge UHID

Expected Results: Selected UHID Merged Successfully with Master UHID Message should be displayed

Score Breakdown: BM25: 100.0% (raw: 68.36) Vector: 93.1% (raw: 0.93) Hybrid: 96.5%

Source: converted-1789826731698.json Created: 9/19/2026

#2 TC_028 Hybrid: 71.5% Found in: both

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Hybrid Search

Combined BM25 + Vector search

Module: Registration P1 Risk: Low Automated

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Registration from the drop down. 4.Functions->Transaction-->Merge/Demerge. 5.Enter Master UHID and Click on Go option. 6.Enter Child UHID and Click on Go option. 7.Click on Merge UHID

Expected Results: Selected UHID Merged Successfully with Master UHID Message should be displayed

Score Breakdown: BM25: 100.0% (raw: 68.36) Vector: 93.1% (raw: 0.93) Hybrid: 96.5%

Source: converted-1789826731698.json Created: 9/19/2026

#2 TC_028 Hybrid: 71.5% Found in: both

Demerge UHID

Module: Registration P1 Risk: Low Automated

Description: Verify user able to Demerge Master UHID and Child UHID.

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Registration from the drop down. 4.Functions->Transaction-->Merge/Demerge. 5.Enter Master UHID and Click on Go option. 6. Click on Demerge Consent and Click on Demerge UHID

Expected Results: Selected UHID Demerged Successfully From Master UHID Message should be displayed

Score Breakdown: BM25: 51.9% (raw: 35.49) Vector: 91.1% (raw: 0.91) Hybrid: 71.5%

Source: converted-1789826731698.json Created: 9/19/2026

Testleaf

RAG Pipeline | Testleaf Enter...

GitHub

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How can I create a new folder...

Ask Gemini

http://localhost:3000

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<RAG Pipeline>Score Fusion

AI semantic reranking & score fusion

AI Semantic Reranking & Score Fusion

Retrieves candidates from MongoDB (BM25 + Vector) and reorders top results using LLM semantic evaluation.

Provider: GROQModel: openai/gpt-oss-120bSynced from .env

Search Query

TC_027 merge UHID records

openai/gpt-oss-120b

Dynamically read from .env; edit anytime

Top-K Candidates

10

Candidate pool

Final Limit

5

Top results

Search

Filters

AI SEMANTIC RERANKING RESULTS: Found 5 test cases evaluated from 75 candidatesModel: GROQ - openai/gpt-oss-120b

Timing: Total: 7889ms | LLM Reranking Time: 3971ms | Method: Hybrid Retrieval → Semantic LLM Scoring

After AI Reranking (Top 5)

Before Reranking (75 Candidates from RAG)

Comparison Table (Top 5 vs 75 Candidates)

After AI Reranking (5 results)

#1 TC_027Rerank Score: 0.9800Was #1 in RAG Candidates

Merge UHID

Testleaf

RAG Pipeline | Testleaf Enter...

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How can I create a new folder...

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<RAG Pipeline>Score Fusion

AI semantic reranking & score fusion

After AI Reranking (Top 5)

Before Reranking (75 Candidates from RAG)

Comparison Table (Top 5 vs 75 Candidates)

After AI Reranking (5 results)

#1 TC_027Rerank Score: 0.9800Was #1 in RAG Candidates

Merge UHID

Module: RegistrationP1Risk: Low

Description: Verify user able to Merge Master UHID and Child UHID.

Ranking Scores:

Vector Score: 0.9310BM25 Score: 25.0138AI Rerank: 0.9800

#2 TC_070Rerank Score: 0.7800Was #26 in RAG Candidates

My Health Care Card

Module: RegistrationP1Risk: Low

Description: Verify User is able to add Family Members under Particular UHID

Ranking Scores:

Vector Score: 0.8842BM25 Score: 0.0000AI Rerank: 0.7800

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DATA PIPELINE

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SEARCH ENGINES

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- Vector Search
- BM25 Search
- Hybrid Search
- Score Fusion**

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RAG Pipeline > Score Fusion

AI semantic reranking & score fusion

#4 TC_071 Rerank Score: 0.7000 Was #27 in RAG Candidates

Search Legacy System Patient Details

Module: Registration P1 Risk: Low

Description: Verify User is able to search Legacy system Data

Ranking Scores:

Vector Score: 0.8840 BM25 Score: 0.0000 AI Rerank: 0.7000

#5 TC_006 Rerank Score: 0.6800 Was #6 in RAG Candidates

OLD to New UHID conversion

Module: Registration P1 Risk: Low

Description: Verify OLD to New UHID conversion

Ranking Scores:

Vector Score: 0.8973 BM25 Score: 9.0565 AI Rerank: 0.6800

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RAG Pipeline > Score Fusion

AI semantic reranking & score fusion

AI SEMANTIC RERANKING RESULTS: Found 5 test cases evaluated from 75 candidates Model: GROQ - openai/gpt-oss-120b
Timing: Total: 7889ms | LLM Reranking Time: 3971ms | Method: Hybrid Retrieval → Semantic LLM Scoring

After AI Reranking (Top 5) Before Reranking (75 Candidates from RAG) Comparison Table (Top 5 vs 75 Candidates)

Initial Candidates (Before AI Reranking - All 75 candidates from RAG)

Showing all 75 candidates retrieved from RAG via hybrid search (BM25 + Vector). Notice their original ranks #1 through #75!

Candidate #1 of 75 TC_027 Vector: 0.9310 BM25: 25.0138 Hybrid Fused: 0.9586

Merge UHID

Module: Registration P1

Description: Verify user able to Merge Master UHID and Child UHID.

Candidate #2 of 75 TC_028 Vector: 0.9106 BM25: 15.3333 Hybrid Fused: 0.7916

Demerge UHID

Module: Registration P1

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RAG Pipeline

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DATA PIPELINE

Convert to JSON

Embeddings & Store

SEARCH ENGINES

Query Preprocessing

Vector Search

BM25 Search

Hybrid Search

Score Fusion

Enterprise AI Stack

MongoDB Atlas • Mistral AI • Groq

RAG Pipeline

Score Fusion

AI semantic reranking & score fusion

After AI Reranking (Top 5)

Before Reranking (75 Candidates from RAG)

Comparison Table (Top 5 vs 75 Candidates)

Before vs After Comparison (Top 5 vs 75 Candidates)

Original Rank (Candidates)	Test Case ID	Title	Original RAG Score	Vector Score	GROQ Rerank Score	Final Rank	Rank Movement
#1 / 75	TC_027	Merge UHID	0.9586	0.9310	0.9800	#1	0
#26 / 75	TC_070	My Health Care Card	0.5305	0.8842	0.7800	#2	+24
#2 / 75	TC_028	Demerge UHID	0.7916	0.9106	0.7000	#3	-1
#27 / 75	TC_071	Search Legacy System Patient Details	0.5304	0.8840	0.7000	#4	+23
#6 / 75	TC_006	OLD to New UHID conversion	0.6832	0.8973	0.6800	#5	+1

How to read this comparison table:

- Original Rank (Candidates):** Position in the 75-candidate pool retrieved from RAG via hybrid search.
- Original RAG Score:** Fused hybrid score combining vector and keyword retrieval.
- Vector Score:** Semantic embedding similarity score (0.0000 - 1.0000).
- GROQ Rerank Score:** Relevance score computed by the AI reranker (openai/gpt-oss-120b).
- Final Rank:** Top-5 reordered position after AI reranking.
- Rank Movement:** +X indicates the item jumped up X positions from deeper in the candidate pool. -Y indicates it moved down.

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Summarize & Dedup

AI summarization & deduplication

TC_027 merge UHID records

Vector

10

Quick Examples:

patient login authentication

merge UHID records

appointment booking

password reset OTP

Search

Deduplicate

Summarize with AI

Copy All Results

Dedup Threshold

0.85

85% (0.85)

Summary Type

Concise

Search Results 10

Deduplicated

AI Summary

Original Results (10)

Copy All

TC_027

Score: 0.9310

Copy

Merge UHID

Verify user able to Merge Master UHID and Child UHID.

Registration

P1 Priority

Integration

TC_028

Score: 0.9106

Copy

Demerge UHID

Verify user able to Demerge Master UHID and Child UHID.

Registration

P1 Priority

Integration

TC_047

Score: 0.8987

Copy

Demerge UHID

Verify user able to Demerge Master UHID and Child UHID.

Registration

P1 Priority

Integration

TC_006

Score: 0.8973

Copy

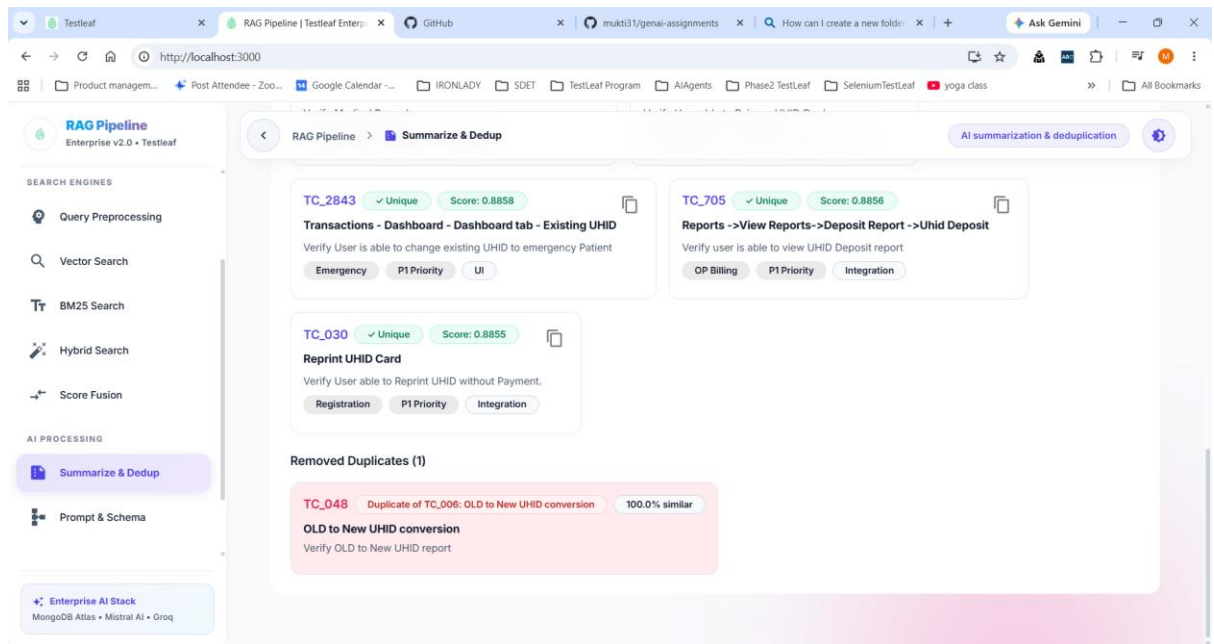
OLD to New UHID conversion

Verify user able to Merge Master UHID and Child UHID.

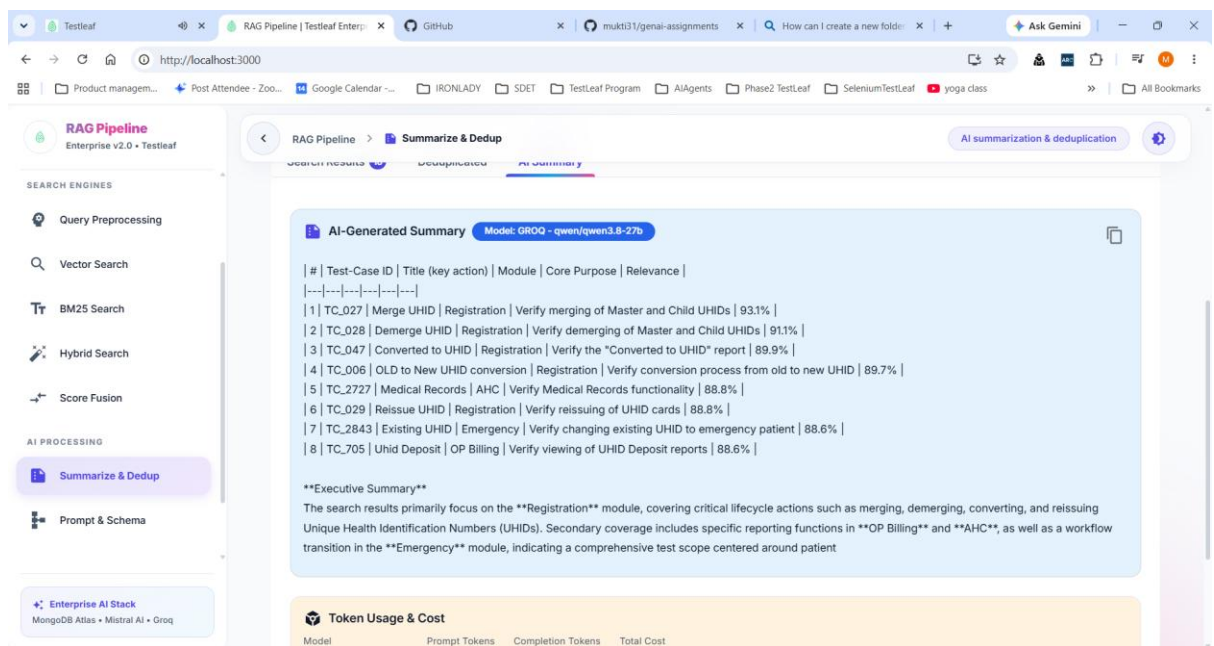
Registration

P1 Priority

Integration



Summarization



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SEARCH ENGINES

- Query Preprocessing
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AI PROCESSING

- Summarize & Dedup
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RAG Pipeline > Summarize & Dedup

AI summarization & deduplication

TC_047 | Converted to UHID | Registration | Verify the "Converted to UHID" report | 89.9% |
TC_006 | OLD to New UHID conversion | Registration | Verify conversion process from old to new UHID | 89.7% |
TC_2727 | Medical Records | AHC | Verify Medical Records functionality | 88.8% |
TC_029 | Reissue UHID | Registration | Verify reissuing of UHID cards | 88.8% |
TC_2843 | Existing UHID | Emergency | Verify changing existing UHID to emergency patient | 88.6% |
TC_705 | Uhid Deposit | OP Billing | Verify viewing of UHID Deposit reports | 88.6% |

****Executive Summary****

The search results primarily focus on the **Registration** module, covering critical lifecycle actions such as merging, demerging, converting, and reissuing Unique Health Identification Numbers (UHIDs). Secondary coverage includes specific reporting functions in **OP Billing** and **AHC**, as well as a workflow transition in the **Emergency** module, indicating a comprehensive test scope centered around patient

Token Usage & Cost

Model	Prompt Tokens	Completion Tokens	Total Cost
qwen/qwen3.8-27b	2,144	330	\$0.000000
	\$0.000000	\$0.000000	2,474 tokens

Model: qwen/qwen3.8-27b

The summary is used for token optimization

Token optimisation is also considered during normalization

End to end pipeline

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SYSTEM CONFIG

Settings

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RAG Pipeline

Prompt & Schema

Configure system prompt templates

Prompt Template & Schema Manager

Configure prompt templates, customize test case counts (Auto vs Custom), and compare With RAG vs Without RAG side-by-side.

Clear Saved Runs

JSON Schema

Prompt Template

Test & Generate

RAG vs Non-RAG Comparison

1/2 Ready

Deept

RAG vs Non-RAG Generated Test Cases Comparison

Contrasting Generated Test Cases (Without RAG) against Generated Test Cases (With RAG) for the exact same user story.

Comparison Requires Both Pipelines to Be Generated

To compare the effect of RAG augmentation, please run generation for both Without RAG (baseline) and With RAG (10-step pipeline).

1. Without RAG (Baseline)

Generated

3 baseline test cases generated directly by LLM.

2. With RAG (Pipeline)

Not Run Yet

Pipeline 10-step MongoDB retrieval pipeline has not been executed yet

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Configure system prompt templates

Generated Test Cases with RAG Pipeline

Mode: With RAGModel: openai/gpt-oss-120b50 MongoDB Matches4,226 / 32000 Tok (13% Budget Safe)Export CSV

Generated Test Cases (3)

Retrieved MongoDB Context (8)

Linked User Stories (10)

Story ID	Title	Module	Priority	Description	Acceptance Criteria	Impact & Risk	Score
HC-285	Radiology Transactions Report Dispatch	Dynamic Forms Management	P2	As a user, I want to radiology transactions report dispatch so that the related functionality works as expected.	Given reports exist in the system, when a user views or generates a report, then the report should display accurate and up-to-date information.	Risk: Low Potentially impacted via shared functionality Same epic/feature area: Dynamic Forms Management	0.557
HC-131	Reports Clinical	Wards - Doctors Management	P2	As a user, I want to reports clinical reports diet so that the related	Given reports exist in the system, when a user views or generates a report, then the	Risk: Low Potentially impacted via shared functionality C...	0.555

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Export CSV

Generated Test Cases (3)

Retrieved MongoDB Context (8)

Linked User Stories (10)

ID	Title	Module	Priority	Type	Preconditions	Test Steps	Ex
						1. Launch the HMS application and login as an authorized diagnostic technician. 2. Navigate to Laboratory → Patient Search and enter the patient's UHID. 3. Select the validated lab report and click 'Share via WhatsApp'. 4. Verify that the system displays the patient's	

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Retrieved MongoDB Context (8)

Linked User Stories (10)

ID	Title	Module	Priority	Preconditions	Test Steps	Expected Results	Score
TC_172	Report Result	Laboratory	P2	N/A	1. Launch the application 2. Login to the application with valid credentials. 3. Click on Laboratory tab from Patient Service Tab. 4. Click Post Analytical Phase-> Report Result 5. Select the Department and Enter LRN/Patient Search No. 6. Click on Search button. 7. Select Test and Enter the values into Parameter Result fields.	User should be able to Report the result and save it	0.556